

## E-Signature Process Overview

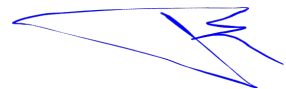
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### Current Flow (Document Preparation)

1. Documents Upload
  2. Recipient Add
  3. Signature Fields
  4. Security
  5. Settings
  6. Review
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### Updated Flow (Document Preparation)

1. Document Upload
2. Choose Path
  - a. Path 1: Self Service (skips Recipient Add)
  - b. Path 2: Recipient Based (follows current flow Steps 2–6)
3. Form Creation (Path 1)
4. Bind Form with Self-Service Envelope
5. Embed Form (on Website or Share Form Link)



### Current Signing Flow

1. Follow the link received via email.
  2. Click on the Sign Field and draw or upload the signature.
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### Upgraded Signing Flow

Step 1: Access the document link via Email, Embedded Website Form, or other source.

Step 2: Pre-Sign Security & Signature Processing

| Step | Action                           | Description / Notes                                               |
|------|----------------------------------|-------------------------------------------------------------------|
| 2.1  | OTP Challenge                    | Email / SMS / Phone verification                                  |
| 2.2  | Prepare Signature Placeholder    | System adds placeholder in PDF and computes exact byte-range hash |
| 2.3  | Generate Cryptographic Signature | X.509 certificate + CMS format                                    |

|     |                                     |                                                                                                           |
|-----|-------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 2.4 | Timestamping (Optional)             | Trusted timestamp token (TSA) added for non-repudiation                                                   |
| 2.5 | Save Signed PDF & Progress Envelope | Envelope moves to next signer in order                                                                    |
| 2.6 | Evidence / Audit Storage            | Store docHash, certSerial, actor, timestamp, OTP result, IP/Device info, optional anchorTx / Merkle proof |
| 2.7 | Validation Endpoint                 | Reconstructs and verifies all stored evidence for compliance                                              |

## Layered Flow of the Enhanced Digital Signature System

| Step                                     | What happens                                            | Libraries involved                                 | Why / Purpose                                                                                                                                  |
|------------------------------------------|---------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Upload Document / Create Envelope</b> | User uploads PDF; empty envelope is created in DB       | None at this step                                  | Existing app logic; sets up envelope and tracks metadata                                                                                       |
| <b>Add Recipients</b>                    | Define recipients, roles, order of signing              | None directly                                      | Existing logic to handle users and permissions                                                                                                 |
| <b>Place Signature Fields</b>            | Map fields (signature, initials, date, etc.) on the PDF | <b>pdf-lib</b>                                     | Prepares the PDF for signing; adds visible signature fields for each recipient                                                                 |
| <b>Send Envelope</b>                     | Email notification sent to recipients                   | None directly                                      | Existing mail service triggers; no crypto needed yet                                                                                           |
| <b>Recipients Sign</b>                   | Recipients click in-field to sign PDF                   | <b>node-signpdf, jsrsasign, speakeasy / otplib</b> | Node-signpdf applies digital signature to PDF; jsrsasign adds timestamp (RFC 3161); speakeasy/otplib handles OTP/TAN for strong authentication |
| <b>Envelope Completed</b>                | All recipients sign → envelope status updated           | <b>winston / pino</b>                              | Write immutable audit trail logs; ensures compliance                                                                                           |

## Crypto Layer (Security & Signing)

| Step                                    | What happens                                 | Libraries                                         | Why                                                                                                           |
|-----------------------------------------|----------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Key &amp; Certificate Generation</b> | Generate RSA/ECDSA keys, X.509 certs         | <b>node-forge, PKI.js</b>                         | Node-forge: create dev/test certs and serial numbers;<br>PKI.js: parsing & validation of certs for compliance |
| <b>PDF Signing</b>                      | Apply signature to PDF                       | <b>node-signpdf</b>                               | Signs the document in a compliant manner (SES/AdES)                                                           |
| <b>Timestamping</b>                     | Record signing time for long-term validation | <b>jsrsasign (TSP), optionally OpenTimestamps</b> | Ensures the signature is tamper-evident and legally valid                                                     |
| <b>OTP/TAN Verification</b>             | Ensure signer is authenticated               | <b>speakeasy / otplib</b>                         | Provides second-factor authentication; satisfies AES compliance in EU/UK and IT Act in India                  |

## Infra Layer (Audit, Hashing, Anchoring)

| Step                        | What happens                                    | Libraries                                 | Why                                                                                |
|-----------------------------|-------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------|
| <b>Hash Document</b>        | Generate SHA-256/512 of signed document         | <b>hasha, Node crypto</b>                 | Creates immutable hash for auditing and anchoring                                  |
| <b>Merkle Root Creation</b> | Batch multiple document hashes                  | <b>merkletreejs</b>                       | Efficient way to anchor multiple documents at once                                 |
| <b>Blockchain Anchoring</b> | Store Merkle root or hash on Ethereum / Bitcoin | <b>ethers.js / web3.js, bitcoinjs-lib</b> | Provides immutable, tamper-proof proof of signing for legal compliance             |
| <b>Audit Trail Logging</b>  | Log all events and signing metadata             | <b>winston / pino</b>                     | Structured, append-only logs to comply with US ESIGN, EU eIDAS, India IT Act, etc. |

## How it differs from your current flow

| Current Flow                                                             | Enhanced Flow with Libraries                                                                                               | Benefits                                                                                                |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Users upload PDF → add recipients → recipients sign → envelope completed | Same basic flow, but <b>PDF signed digitally, timestamps added, hashes anchored, OTP verification</b> , audit logs written | Legal compliance (SES, AdES, AES), tamper-proof signatures, stronger authentication, future LTV support |